Question 3 **[6 Marks]**

How many words, with or without meaning can be made from letters of the word TUESDAY, assuming that no letter is repeated, if:

a) 4 letters are used at a time. (2 marks)

b) All letters are used at a time. (2 marks)

c) All letters are used but first letter is a vowel. (2 marks)

a): "TUESDAY" ... has 7 unique letters

Formula $P(n,r) = \frac{n!}{(n-r)!}$ $n=7$ $r=4$

Formula $P(7,4) = \frac{7!}{(7-4)!} = \boxed{840}$ Answer

b):

"TUESDAY" ... has 7 unique letters, We will arrange all letters

Formula $7! = \boxed{5040}$ Answer

c): Vowels: U, E, A

First letters = 3 choices (one Vowel)

Remaining 6 letters can be arranged in 6! ways

Formula: Total Arrangements = (Ways to Choose First Letter) × (Arrangements of Remaining Letters)

$= 3 \times 6! = 3 \times 720 = \boxed{2160}$

Question 4 **[9 Marks]**

A committee of 7 must be formed from 9 boys and 4 girls. In how many ways this can be done when the committee consist of:

a) Exactly 3 girls. (2 marks)

b) At least 3 girls. (3 marks)

c) At most 3 girls. (4 marks)

a): Total = 9 boys, 4 girls

we select 3 girls from 4 = $\binom{4}{3} = \frac{4!}{3!(4-3)!} = 4$

we select 4 boys from 9 = $\binom{9}{4} = \frac{9\times8\times7\times6}{4\times3\times2\times1} = 126$

Total Ways: $\binom{4}{3} \times \binom{9}{4} = 4 \times 126 = \boxed{504}$ Answer

b): Possible Cases: 4G+3B OR 3G+4B

- Case (1): 3G+4B
  from Part(a) = 504
- Case (2): 4G+4B
  $\binom{4}{4}=1$, $\binom{9}{3} = \frac{9\times8\times7}{3\times2\times1} = 84$

Total Ways:
504 + (1 × 84)
$= \boxed{588}$ Answer

c): girls(G) boys(B)

Possible Cases: 0G+7B + 1G+6B + 2G+5B + 3G+4B

- Case (1): 0G+7B
  $\binom{4}{0} \times \binom{9}{7} = 1 \times 36 = 36$
- Case (2): 1G+6B
  $\binom{4}{1} \times \binom{9}{6} = 4 \times 84 = 336$
- Case (3): 2G+5B
  $\binom{4}{2} \times \binom{9}{5} = 6 \times 126 = 759$
- Case (4): 3G+4B
  from Part(a) = 504

Total Ways:
36 + 336 + 759 + 504
$= \boxed{1632}$ Answer